

ANSWER KEY

GATE CS — GATE CS 2026 Trend-Based Mock Paper
2

Q.No.	Type	Answer	Marks
1	2MSQ	—	2
2	2NAT	B	2
3	2NAT	—	2
4	2NAT	—	2
5	2MCQ	—	2
6	2MCQ	—	2
7	2NAT	A	2
8	2NAT	—	2
9	2MCQ	B	2
10	2MCQ	C	2
11	2MCQ	C	2
12	2NAT	B	2
13	2NAT	B	2
14	2MCQ	B	2
15	2MSQ	C	2
16	2NAT	A	2
17	1NAT	B	1
18	1MCQ	C	1
19	1NAT	B	1
20	1MCQ	C	1
21	1MCQ	B	1
22	2MCQ	C	2
23	1MCQ	B	1
24	1NAT	C	1
25	2NAT	C	2
26	1MCQ	C	1
27	1MCQ	B	1
28	1MCQ	A	1
29	1NAT	B	1
30	1MCQ	A	1
31	1MCQ	B	1
32	1MCQ	C	1
33	1NAT	B	1
34	1MCQ	B	1
35	1NAT	B	1
36	1MCQ	B	1

37	1NAT	C	1
38	1MCQ	A	1
39	1NAT	B	1
40	1MCQ	B	1
41	1MCQ	B	1
42	1MCQ	B	1
43	1MCQ	B	1
44	1MCQ	A	1
45	2MCQ	—	2
46	1MCQ	A	1
47	2MCQ	—	2
48	1MCQ	A	1
49	2MSQ	—	2
50	2MCQ	—	2
51	2MCQ	—	2
52	2NAT	—	2
53	2MCQ	—	2
54	2MCQ	—	2
55	2MCQ	—	2
56	1MCQ	—	1
57	2NAT	—	2
58	2MCQ	—	2
59	2NAT	—	2
60	2NAT	—	2
61	1MCQ	—	1
62	2MCQ	—	2
63	2NAT	—	2
64	2NAT	—	2
65	2NAT	—	2

Q1. General Aptitude | 2 marks | 2MSQ

Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man. Select ...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q1

Q2. General Aptitude | 2 marks | 2NAT

Antonym of 'ephemeral':

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q10

Q3. General Aptitude | 2 marks | 2NAT

Bird : Nest :: Bee : _____ Select the correct option to complete the analogy.

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q2

Q4. General Aptitude | 2 marks | 2NAT

For positive non-zero real variables and , if $\ln + = 1 [\ln() + \ln()] (2) 2$ then, the value of + is

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q4

Q5. General Aptitude | 2 marks | 2MCQ

Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered. Select the...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q1

Q6. General Aptitude | 2 marks | 2MCQ

If `` denotes increasing order of intensity, then the meaning of the words [walk jog sprint] is analogous to [bothered _...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q1

Q7. General Aptitude | 2 marks | 2NAT

In a class of 60 students, 30 play cricket, 25 play football, and 15 play both. How many play neither?

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 CS Session 1 Q10

Q8. General Aptitude | 2 marks | 2NAT

In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of s...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q3

Q9. Engineering Mathematics | 2 marks | 2MCQ

Determinant of 2x2 identity:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q1

Q10. Engineering Mathematics | 2 marks | 2MCQ
Probability of at least one head in 3 coin tosses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q11

Q11. Engineering Mathematics | 2 marks | 2MCQ
Let A be a 3×3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q1

Q12. Engineering Mathematics | 2 marks | 2NAT
Mean of binomial $n=10, p=0.5$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q1

Q13. Engineering Mathematics | 2 marks | 2NAT
Derivative of x^3 at $x=2$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q1

Q14. Programming and Data Structures | 2 marks | 2MCQ
Infix to postfix uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q6

Q15. Programming and Data Structures | 2 marks | 2MSQ
Avg hash lookup:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q3

Q16. Programming and Data Structures | 2 marks | 2NAT
Inorder traversal of BST gives:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q3

Q17. Programming and Data Structures | 1 marks | 1NAT
What is the time complexity of inserting at the beginning of a linked list?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q3

Q18. Algorithms | 1 marks | 1MCQ
Kruskal uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q2

Q19. Algorithms | 1 marks | 1NAT
Worst-case time complexity of merge sort:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q12

Q20. Algorithms | 1 marks | 1MCQ
LCS uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2023 CS Session 1 Q7

Q21. Algorithms | 1 marks | 1MCQ
Time complexity of binary search:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q11

Q22. Algorithms | 2 marks | 2MCQ
The time complexity of Dijkstra's algorithm using a binary heap is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q7

Q23. Algorithms | 1 marks | 1MCQ
BFS uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q7

Q24. Algorithms | 1 marks | 1NAT
Merge sort space:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q7

Q25. Operating Systems | 2 marks | 2NAT
*Using 3 frames, which page replacement algorithm would not cause any page fault for reference string
0,1,2,3,0,1,4,0,1,2...*

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q8

Q26. Operating Systems | 1 marks | 1MCQ
Which of the following is NOT a necessary condition for deadlock?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q4

Q27. Operating Systems | 1 marks | 1MCQ
Semaphore with value 1 is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q8

Q28. Operating Systems | 1 marks | 1MCQ
TLB stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 CS Session 1 Q9

Q29. Operating Systems | 1 marks | 1NAT
Banker's algorithm is for:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q4

Q30. Operating Systems | 1 marks | 1MCQ
Round-robin uses:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q4

Q31. Operating Systems | 1 marks | 1MCQ
Inode contains:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q4

Q32. Computer Networks | 1 marks | 1MCQ
Which protocol operates at the application layer of the TCP/IP model?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q6

Q33. Computer Networks | 1 marks | 1NAT
CSMA/CD is used in:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q13

Q34. Computer Networks | 1 marks | 1MCQ
OSPF is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q6

Q35. Computer Networks | 1 marks | 1NAT
Protocol to find MAC from IP:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q6

Q36. Computer Networks | 1 marks | 1MCQ
TCP connection establishment uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q14

Q37. Computer Networks | 1 marks | 1NAT
Asymmetric encryption:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q5

Q38. Digital Logic | 1 marks | 1MCQ
Flip-flop stores:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 CS Session 1 Q2

Q39. Digital Logic | 1 marks | 1NAT
Full adder has how many inputs:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q2

Q40. Digital Logic | 1 marks | 1MCQ
The number of Boolean functions of n variables is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q2

Q41. Theory of Computation | 1 marks | 1MCQ
Which language is regular?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q13

Q42. Theory of Computation | 1 marks | 1MCQ
PDA recognizes:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q14

Q43. Theory of Computation | 1 marks | 1MCQ
Halting problem:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q8

Q44. Compiler Design | 1 marks | 1MCQ
Lexical analyzer outputs:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q15

Q45. CS | 2 marks | 2MCQ
Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a ...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 1 Q65

Q46. Compiler Design | 1 marks | 1MCQ
Lexical analyzer outputs:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q15

Q47. CS | 2 marks | 2MCQ
A person sold two different items at the same price. He made 10% profit in one item, and 10% loss in the other item. In ...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q7

Q48. Digital Logic | 1 marks | 1MCQ
Flip-flop stores:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 CS Session 1 Q2

Q49. CS | 2 marks | 2MSQ
Consider the given sequential circuit designed using D-Flip-flops. The circuit is initialized with some value (initial s...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q60

Q50. CS | 2 marks | 2MCQ
Consider the following code snippet using the fork() and wait() system calls. Assume that the code compiles and runs correctly...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 1 Q57

Q51. CS | 2 marks | 2MCQ

Let $R1$ and $R2$ be two 4-bit registers that store numbers in 2's complement form. For the operation $R1+R2$, which one of the...

Answer: See explanation

Source: Adapted from GATE 2022 CS Session 1 Q18

Q52. CS | 2 marks | 2NAT

Consider the following C program: `#include <stdio.h> int gate (int n) { int d, t, newnum, turn; newnum = turn = 0; t=1; ...`

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q63

Q53. CS | 2 marks | 2MCQ

The pie charts depict the shares of various power generation technologies in the total electricity generation of a count...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q8

Q54. CS | 2 marks | 2MCQ

Suppose we are given n keys, m hash table slots, and two simple uniform hash functions $h1$ and $h2$. Further suppose our h...

Answer: See explanation

Source: Adapted from GATE 2022 CS Session 1 Q16

Q55. CS | 2 marks | 2MCQ

Consider a process P running on a CPU. Which one or more of the following events will always trigger a context switch by...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q25

Q56. General Aptitude | 1 marks | 1MCQ

Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered. Select the...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q1

Q57. CS | 2 marks | 2NAT

$= \{0, 1, 2, 3, \dots\}$ is the set of non-negative integers. Let be the set of functions from to itself. For any two functi...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q49

Q58. CS | 2 marks | 2MCQ

The following two signed 2's complement numbers (multiplicand M and multiplier Q) are being multiplied using Booth's alg...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q32

Q59. CS | 2 marks | 2NAT

Sequence the following sentences in a coherent passage. P : This fortuitous geological event generated a colossal amount ...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q6

Q60. CS | 2 marks | 2NAT
The value of the following limit is _____. $\lim_{x \rightarrow 0} \frac{1}{e^2 x}$

Answer: See explanation

Source: Adapted from GATE 2022 CS Session 1 Q34

Q61. General Aptitude | 1 marks | 1MCQ
If $e^x = -1$ for all real values of x , which one of the following statements is true?

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q3

Q62. CS | 2 marks | 2MCQ
Let G be an edge-weighted undirected graph with positive edge weights. Suppose a positive constant is added to the weight of every edge in G .

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q37

Q63. CS | 2 marks | 2NAT
Consider the following recurrence relation: $T(n) = T(n-1) + 1$, $T(1) = 1$. Which one of the following options is CORRECT?

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 1 Q42

Q64. CS | 2 marks | 2NAT
Let A be a priority queue for maintaining a set of elements. Suppose A is implemented using a max-heap data structure. Then,

Answer: See explanation

Source: Adapted from GATE 2023 CS Session 1 Q46

Q65. CS | 2 marks | 2NAT
Consider the operator precedence and associativity rules for the integer arithmetic operators given in the table below.

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 1 Q33